

dogo_patterns

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Pattern Mining with daltoolbox

Descriptive learners that discover structure directly from co-occurrence data — a family that asks a different didactic question than classification and regression: how do we configure the search space, constrain the discovered patterns, and summarize their quality in a reproducible way?

4 TECHNIQUES • PAT_APRIORI • PAT_ECLAT • PAT_CSPADE • DARA / INTEREST FILTERS

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Navigating Pattern Mining

Four progressively different questions about the same transactional data:

1. **Association Rules** — directional patterns with a left-hand side implying a right-hand side.
2. **Itemsets** — undirected co-occurrence structures, filtered by which items may appear.
3. **Sequential Patterns** — recurring structures where *order* matters, not just co-occurrence.

4. **Interesting Rule Filters** — post-mining filters that decide which discovered rules are worth keeping.

A shared mechanic runs through all of them: when `supp`, `conf`, or `support` are set to `0`, the miner estimates the threshold during `fit()` through explicit strategies such as `pat_support_threshold()` and `pat_confidence_threshold()`.

Association Rules — `pat_apriori`

Directional rules with a fixed consequent (`rhs`) and explicit filtering by confidence and lift — the entry point for the whole pattern-mining family.

```
pm <- pat_apriori(  
  target = "rules", supp = 0, conf = 0,  
  support_strategy = pat_support_threshold("curvature"),  
  confidence_strategy = pat_confidence_threshold("rhs_baseline", margin =  
    0.1),  
  minlen = 2, maxlen = 3,  
  rhs = c("native-country=United-States"),  
  quality_filter = patutils()$quality_min(confidence = 0.9, lift = 1.03)  
)  
pm <- fit(pm, trans)  
rules <- discover(pm, trans)
```

- A fixed `rhs` lets `pat_confidence_threshold("rhs_baseline")` set confidence relative to how frequent the consequent already is.
- `lift` acts as a second filter on top of support/confidence — it removes rules that are frequent but not informative.

Itemsets — pat_eclat

Frequent itemset mining: groups of items that often appear together, without any directional implication.

```
pm <- pat_eclat(  
  supp = 0, support_strategy = pat_support_threshold("curvature"),  
  maxlen = 3,  
  include = c("sex=Male", "marital-status=Married-civ-spouse",  
    "race=White"),  
  exclude = c("income=small"),  
  quality_filter = patutils()$quality_min(support = 0.4)  
)  
pm <- fit(pm, trans)  
itemsets <- discover(pm, trans)
```

- `include` / `exclude` act as semantic filters over which items may appear in the result — a structural filter, applied before the quality filter.
- Itemsets capture co-occurrence, not implication — the key conceptual difference from association rules.

Sequential Patterns — pat_cspade

The same pattern-mining workflow, but the data type shifts from itemsets to ordered event sequences — the analytical question becomes "what recurring order of events happens," not "what items co-occur."

```
pm <- pat_cspade(  
  support = 0,  
  support_strategy = pat_support_threshold("min_count", min_count = 4),  
  maxlen = 3,  
  quality_filter = patutils()$quality_min(support = 0.5)  
)  
pm <- fit(pm, sequence_data)  
seqs <- discover(pm, sequence_data)
```

- Input is sequence-transaction data (each case is an ordered basket of events), not a flat transaction table.
- The workflow is otherwise unchanged from `pat_apriori` / `pat_eclat` — same configure/fit/discover/evaluate rhythm.

Interesting Rule Filters

Discovery and interestingness filtering are separate concerns: a miner generates candidate rules, and a filter decides which ones remain.

- `pat_rule_filter_none()` — the explicit no-filter default, keeps every candidate rule.
- `pat_rule_filter_interest(lift_min, kulc_min, ir_max)` — classical measures computed directly from support relationships (lift, Kulczynski, imbalance ratio).
- `pat_rule_filter_dara(...)` — Divergent Association Ranking Analysis: scores each antecedent by how much more prominent it becomes in the rule set than in the target subset, and can be inspected as an explanatory ranking (`dara_ranking`).
- `pat_filter_rules(rules, filter)` — applies any of the above to a rule table or an `arules` rule object; a `rule_filter` can also be passed directly into `pat_apriori()` .

```
dara_filter <- pat_rule_filter_dara(data, rhs_attr = "income", rhs_value =  
  "small", min_score = 1)  
dara_rules <- pat_filter_rules(rules, dara_filter)  
attr(dara_rules, "dara_ranking")[["marital-status"]]
```

What to Observe

- All four techniques share one experiment shape: configure the miner → `fit()` → `discover()` → `evaluate()` .
- The real design decisions live in the thresholds and filters, not in the workflow — `supp` / `conf` strategies control the search space, `include` / `exclude` control the vocabulary, and interest filters control what survives after mining.